

CSC 1300 PROGRAM 4

"Jeopardy..."



Concepts

- Pointers
- Dynamic memory allocation
- Using multiple files – header file, source files, text files
- Arrays
- Using text from files

Description

You've heard of Jeopardy! (I hope), now get read for "Jeopardy...". This game is very similar to the beloved classic, where you will have categories with different questions worth different point values. If the contestant can answer correctly, they will get the points for that question. Otherwise, those very points will be deducted. Unfortunately, due to legal reasons, there need to be some slight differences in your game such as contestants can't steal questions, daily doubles are changed somewhat, and more to make this just slightly less fun!

What You Can't Use in This Program

You may not use code or syntax in your programming assignment that has not been taught in the class without referencing your source.

You may not use the following in your programs:

- while(true) or while(1)

- goto statements
- recursion including recursively calling main
- exit()
- ternary operators (any syntax requiring a '?')
- lambda functions
- try-catch
- generative AI (see below)

You may not use generative AI (genAI) in any way for this program assignment.

Files

Given Files

- **sample category questions** – including science.txt, history.txt, math.txt, music.txt, comic.txt
- **functions.cpp** – the functions for this program have been partially implemented (just so you don't have to use setw because we like you)

Files that MUST be included in your submission

- **jeopardy.h** – contains all includes, namespace std, and function prototypes for functions defined in functions.cpp (also needs to #include "structs.h")
- **prog4.cpp** – contains the main function
- **functions.cpp** – contains all the programmer-defined functions
- ***.txt** – if you create new categories/questions, include the .txt file(s) with your submission
- **Prog4reportproof**

Specifications

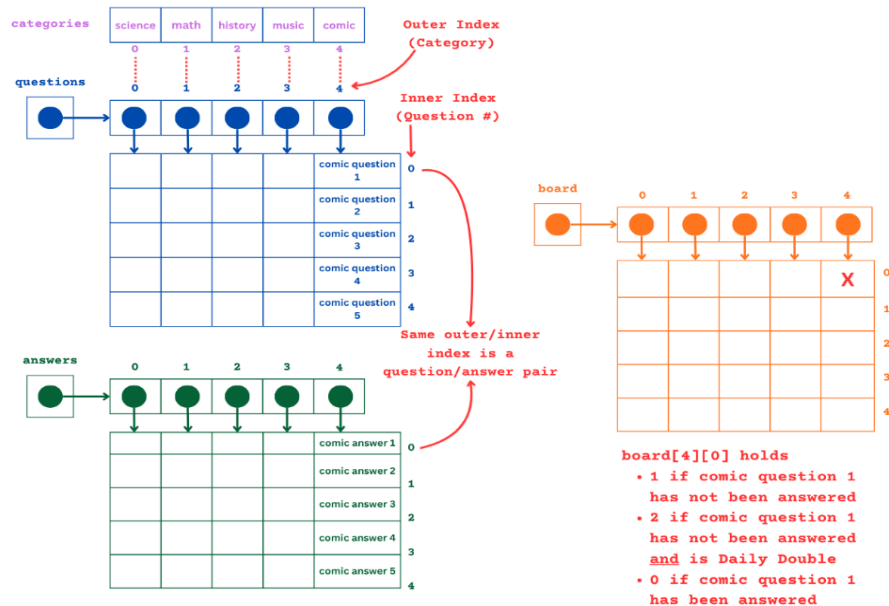
- Use good commenting/documentation practices as taught in the course.
- Code should be readable with consistent indentions.

Program 4 Header File – jeopardy.h

The header file **jeopardy.h** should contain all includes and function prototypes. The file should also include a top comment block and sufficient inline comments.

Main Function – prog4.cpp

- Your **prog4.cpp** file must **#include "jeopardy.h"**
- The main function should have 3 constant size variables – one for the number of users (which should be 3), one for the number of categories (which should be at least 5), and one should be for the number of questions (which should be 5).
- You should have a categories array to hold the name of each category.
- You will have **3 pointers to an array of pointers**. One to hold the questions, one to hold the answers, and one for the board status. All the arrays should be dynamically allocated. The image below show how each array relates to one another.



- You should also have an array to hold the user's names and the user's points.
- Main should then call the loadQuestionsAndAnswers() function, the initializeBoard() function, and the printWelcomeMessage() function.
- Then, the program should take in each of the player's names.
- The game loop should continue as long as there are more questions. In the game loop, it should give each player a turn. The game loop will have the following:
 - It should call the showBoard() function to display the board and check if there are available questions
 - It should allow the user to enter in their choice for category and question. **All user input should be validated as shown in the sample output.**
 - The program should check if the user's question is a Daily Double and print a message if it is.
 - The program should call printQuestionGetAnswer() to print the user's selected question and get the answer.
 - The answer should then be passed to checkAnswer()
 - If the user answered the question correctly, a message should be printed and points added to the user's total. If the question is a Daily Double, double the amount of points should be added (so a 100 should add 200, a 200 should add 400, etc).
 - If the user answered incorrectly, the points should be subtracted from the total. If the question was a Daily Double, you do not need to subtract double points.
 - At the end of each round (all 3 players have had a turn), the program should print out the current standings.
- After the game, the program should call getWinner() and print a message for the winner.
- At the end of main, all dynamically allocated memory should be released.

Programmer-Defined Functions – functions.cpp

- Your `functions.cpp` file must must `#include "jeopardy.h"`
- The functions that should be in this source file are described below

`bool showBoard(string[], int**, int, int)`

- This function is given
- This function will take the categories, board status, number of categories, and number of questions and print the current board for the user

`void printWelcomeMessage()`

- This function is given
- This will print the game title

`void loadQuestionsAndAnswers(string[], string**, string**, int, int)`

- This function should take in the categories array, questions array, answers array, number of categories, and the number of questions
- For each category, that category's question file should be opened (name category_name.txt).
- The file should be laid out as Question#Answer#Points
- The question and answer should be pulled from the file. Then, the points should be used to determine where that question and answer would be put.
- If a question/answer with that point value in that category has already been filled, it should skip that line of the file.
- Otherwise, the question and answer should be put in the correct spot of the question and answers arrays respectively.
- Note: If you have a Mac/Linux the file names will be case sensitive (so science.txt is not the same as Science.txt). Windows is not case sensitive.

`void intializeBoard(int**, int, int)`

- This function should take in the board status, number of categories, and number of questions
- Each spot in the board array should be initialized to 1.
- There should be 3 random indexes (category and question) chosen to be Daily Doubles. These spots should be set to 2.

`int getWinner(int[], int)`

- This function should take in an array with the points for each user and the number of users
- It should return the index of the user with the most points

`string printQuestionsGetAnswer(string**, int**, int, int)`

- This function should take in the questions array, the board status, the users selected category, and the users selected question number
- It should print the correct question
- It should update the board status to 0 for the question the user chose as that question is no longer available
- It should return the user's answer input

`string toLowerString(string)`

- This function should take in a string
- It will change each character to lowercase and return the new lowercase string

`bool checkAnswer(string**, string, int, int)`

- This function should take in the answers array, the users given answer, the selected category, and the selected question num
- The user answer and the correct answer should both be changed to all lowercase

- The function should return true if the user answer matches the correct answer exactly or if the correct answer is within the user answer (so “What is blue?” contains “blue” and is still a correct answer).

Program Report

Fill out this program report and take a screenshot of the confirmation page after submitting and name it **Prog4ReportProof**.

Program Report Link: https://tntech.co1.qualtrics.com/jfe/form/SV_5vhCYdoB1bhJXM2

Sample Output

User input is highlighted in yellow.

```
*****
*
*                               *
*           Welcome to "Jeopardy..."           *
*
*****

Player 1, what is your name? Cal Stewart
Player 2, what is your name? Sara Owns
Player 3, what is your name? Lela Gracy

===== NEW ROUND =====

      | 1 | 2 | 3 | 4 | 5 |
-----
1. Science    | 100 | 200 | 300 | 400 | 500 |
2. History    | 100 | 200 | 300 | 400 | 500 |
3. Math       | 100 | 200 | 300 | 400 | 500 |
4. Comics     | 100 | 200 | 300 | 400 | 500 |
5. Music      | 100 | 200 | 300 | 400 | 500 |
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Cal Stewart, it's your turn!

Select a category (1-5): science
Invalid input. Please select a category (1-5): 9
Invalid input. Please select a category (1-5): 1
Select a question (1-5): 500
Invalid input. Please select a question (1-5): five hundred
Invalid input. Please select a question (1-5): 5

Question: What theory explains space time and gravity?
Your answer: What is the theory of relativity?

Correct!

      | 1 | 2 | 3 | 4 | 5 |
-----
1. Science    | 100 | 200 | 300 | 400 | X |
2. History    | 100 | 200 | 300 | 400 | 500 |
3. Math       | 100 | 200 | 300 | 400 | 500 |
4. Comics     | 100 | 200 | 300 | 400 | 500 |
5. Music      | 100 | 200 | 300 | 400 | 500 |
-----
-----
```

Sara Owns, it's your turn!

Select a category (1-5): 3

Select a question (1-5): 3

Question: What operation undoes multiplication?

Your answer: division

Correct!

	1	2	3	4	5
1. Science	100	200	300	400	X
2. History	100	200	300	400	500
3. Math	100	200	X	400	500
4. Comics	100	200	300	400	500
5. Music	100	200	300	400	500

Lela Gracy, it's your turn!

Select a category (1-5): 2

Select a question (1-5): 5

Question: What was a key result of industrialization?

Your answer: 8 hour work day, mass produced products and manufactured goods, trade expanded, people found jobs, more women had right for wages

Incorrect. The correct answer was: urbanization

*****Current Points*****

Cal Stewart : 500 points

Sara Owns : 300 points

Lela Gracy : -500 points

===== NEW ROUND =====

	1	2	3	4	5
1. Science	100	200	300	400	X
2. History	100	200	300	400	X
3. Math	100	200	X	400	500
4. Comics	100	200	300	400	500
5. Music	100	200	300	400	500

Cal Stewart, it's your turn!

Select a category (1-5): 1

Select a question (1-5): 4

Question: What law states that an object in motion stays in motion?

Your answer: One of Newton's 3

Incorrect. The correct answer was: first law

	1	2	3	4	5
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1. Science		100		200		300		X		X	
2. History		100		200		300		400		X	
3. Math		100		200		X		400		500	
4. Comics		100		200		300		400		500	
5. Music		100		200		300		400		500	

Sara Owns, it's your turn!

Select a category (1-5): 3

Select a question (1-5): 5

Question: What is the measure of instantaneous rate of change called?

Your answer: jerk

Incorrect. The correct answer was: derivative

		1		2		3		4		5	
1. Science		100		200		300		X		X	
2. History		100		200		300		400		X	
3. Math		100		200		X		400		X	
4. Comics		100		200		300		400		500	
5. Music		100		200		300		400		500	

Lela Gracy, it's your turn!

Select a category (1-5): 1

Select a question (1-5): 1

Question: What state of matter has a definite shape and volume?

Your answer: What is a solid?

Correct!

*****Current Points*****

Cal Stewart : 100 points

Sara Owns : -200 points

Lela Gracy : -400 points

===== NEW ROUND =====

		1		2		3		4		5	
1. Science		X		200		300		X		X	
2. History		100		200		300		400		X	
3. Math		100		200		X		400		X	
4. Comics		100		200		300		400		500	
5. Music		100		200		300		400		500	

Cal Stewart, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 3

Question: What is the name of the Marvel superhero team?

Your answer: The Avengers

Correct!

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	400	X
3. Math	100	200	X	400	X
4. Comics	100	200	X	400	500
5. Music	100	200	300	400	500

Sara Owns, it's your turn!

Select a category (1-5): 5

Select a question (1-5): 3

Question: What is the main tune of a song called?

Your answer: The melody

Correct!

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	400	X
3. Math	100	200	X	400	X
4. Comics	100	200	X	400	500
5. Music	100	200	X	400	500

Lela Gracy, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 1

*** DAILY DOUBLE! ***

Question: What is the real name of Batman?

Your answer: Bruce Wayne

Correct!

You found a DAILY DOUBLE! Points doubled!

*****Current Points*****

Cal Stewart : 400 points

Sara Owns : 100 points

Lela Gracy : -200 points

===== NEW ROUND =====

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	400	X
3. Math	100	200	X	400	X

4. Comics		X		200		X		400		500	
5. Music		100		200		X		400		500	

Cal Stewart, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 4

*** DAILY DOUBLE! ***

Question: What is a story involving multiple series called?

Your answer: The multiverse

Incorrect. The correct answer was: crossover

		1		2		3		4		5	

1. Science		X		200		300		X		X	
2. History		100		200		300		400		X	
3. Math		100		200		X		400		X	
4. Comics		X		200		X		X		500	
5. Music		100		200		X		400		500	

Sara Owns, it's your turn!

Select a category (1-5): 5

Select a question (1-5): 5

Question: What set of rules guides musical composition?

Your answer: Music theory

Correct!

		1		2		3		4		5	

1. Science		X		200		300		X		X	
2. History		100		200		300		400		X	
3. Math		100		200		X		400		X	
4. Comics		X		200		X		X		500	
5. Music		100		200		X		400		X	

Lela Gracy, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 5

Question: What quality makes a superhero interesting?

Your answer: depth

Incorrect. The correct answer was: character depth

*****Current Points*****

Cal Stewart : 0 points

Sara Owns : 600 points

Lela Gracy : -700 points

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===== NEW ROUND =====

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	400	X
3. Math	100	200	X	400	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	400	X

Cal Stewart, it's your turn!

Select a category (1-5): 2

Select a question (1-5): 4

Question: What was a major reason for the fall of Rome?

Your answer: expansion

Incorrect. The correct answer was: instability

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	X	X
3. Math	100	200	X	400	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	400	X

Sara Owns, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 4

0_0

We have to do that all over again, you fool

Select a category (1-5): 5

Select a question (1-5): 4

Question: What is the structure of a musical piece called?

Your answer: Its composition

Incorrect. The correct answer was: form

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	X	X
3. Math	100	200	X	400	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	X	X

Lela Gracy, it's your turn!

Select a category (1-5): 3

Select a question (1-5): 4

Question: What is the point where a graph crosses the y axis called?

Your answer: the y intercept?

Correct!

*****Current Points*****

Cal Stewart : -400 points
Sara Owns : 200 points
Lela Gracy : -300 points

=====

===== NEW ROUND =====

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	300	X	X
3. Math	100	200	X	X	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	X	X

Cal Stewart, it's your turn!

Select a category (1-5): 2

Select a question (1-5): 3

Question: What event triggered World War I?

Your answer: The assassination of Archduke Franz Ferdinand

Correct!

	1	2	3	4	5
1. Science	X	200	300	X	X
2. History	100	200	X	X	X
3. Math	100	200	X	X	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	X	X

Sara Owns, it's your turn!

Select a category (1-5): 1

Select a question (1-5): 3

Question: What is speed with direction called?

Your answer: is it velocity???

Correct!

	1	2	3	4	5
1. Science	X	200	X	X	X
2. History	100	200	X	X	X
3. Math	100	200	X	X	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	X	X

Lela Gracy, it's your turn!

Select a category (1-5): 1

Select a question (1-5): 2

Question: What is the ability to do work called?

Your answer: force? i dont- ENERGY

Correct!

*****Current Points*****

Cal Stewart : -100 points
Sara Owns : 500 points
Lela Gracy : -100 points

===== NEW ROUND =====

	1	2	3	4	5
1. Science	X	X	X	X	X
2. History	100	200	X	X	X
3. Math	100	200	X	X	X
4. Comics	X	200	X	X	X
5. Music	100	200	X	X	X

Cal Stewart, it's your turn!

Select a category (1-5): 4

Select a question (1-5): 2

Question: What city does Batman protect?

Your answer: Gotham City

Correct!

	1	2	3	4	5
1. Science	X	X	X	X	X
2. History	100	200	X	X	X
3. Math	100	200	X	X	X
4. Comics	X	X	X	X	X
5. Music	100	200	X	X	X

Sara Owns, it's your turn!

Select a category (1-5): 5

Select a question (1-5): 2

Question: What is the speed of music called?

Your answer: tempo

Correct!

	1	2	3	4	5
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1. Science		X		X		X		X		X	
2. History		100		200		X		X		X	
3. Math		100		200		X		X		X	
4. Comics		X		X		X		X		X	
5. Music		100		X		X		X		X	

Lela Gracy, it's your turn!

Select a category (1-5): 3

Select a question (1-5): 2

Question: What is used to control the order of operations in expressions?

Your answer: PEMDAS

Incorrect. The correct answer was: parentheses

*****Current Points*****

Cal Stewart : 100 points
Sara Owns : 700 points
Lela Gracy : -300 points

===== NEW ROUND =====

		1		2		3		4		5	

1. Science		X		X		X		X		X	
2. History		100		200		X		X		X	
3. Math		100		X		X		X		X	
4. Comics		X		X		X		X		X	
5. Music		100		X		X		X		X	

Cal Stewart, it's your turn!

Select a category (1-5): 2

Select a question (1-5): 2

Question: What was the conflict between the US and USSR called?

Your answer: The Cold War

Correct!

		1		2		3		4		5	

1. Science		X		X		X		X		X	
2. History		100		X		X		X		X	
3. Math		100		X		X		X		X	
4. Comics		X		X		X		X		X	
5. Music		100		X		X		X		X	

Sara Owns, it's your turn!

Select a category (1-5): 5

Select a question (1-5): 1

*** DAILY DOUBLE! ***

Question: What is a single musical sound called?

Your answer: note

Correct!

You found a DAILY DOUBLE! Points doubled!

	1	2	3	4	5
1. Science	X	X	X	X	X
2. History	100	X	X	X	X
3. Math	100	X	X	X	X
4. Comics	X	X	X	X	X
5. Music	X	X	X	X	X

Lela Gracy, it's your turn!

Select a category (1-5): 1

Select a question (1-5): 1

0 0

We have to do that all over again, you fool

Select a category (1-5): 2

Select a question (1-5): 1

Question: Who was a leader in the civil rights movement?

Your answer: Martin Luther King Jr (MLK)

Correct!

*****Current Points*****

Cal Stewart : 300 points

Sara Owns : 900 points

Lela Gracy : -200 points

===== NEW ROUND =====

	1	2	3	4	5
1. Science	X	X	X	X	X
2. History	X	X	X	X	X
3. Math	100	X	X	X	X
4. Comics	X	X	X	X	X
5. Music	X	X	X	X	X

Cal Stewart, it's your turn!

Select a category (1-5): 3

Select a question (1-5): 1

Question: What is the rate of change between two variables called?

Your answer: slope

Correct!

	1	2	3	4	5
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1. Science		X		X		X		X		X	
2. History		X		X		X		X		X	
3. Math		X		X		X		X		X	
4. Comics		X		X		X		X		X	
5. Music		X		X		X		X		X	

*****Current Points*****

Cal Stewart : 400 points
Sara Owns : 900 points
Lela Gracy : -200 points

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Congratulations Sara Owns! You are the winner with 900 points!